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Class 7 Math Notes In English Chapter – 07 Congruence of Triangles

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Congruence of Triangles

1. Two figures having exactly same shape and size are said to be congruent.

2. Two plane figures are congruent, if each one when superimposed on the other, covers the other exactly.

3. Congruence of some geometrical figures:

i. Two line segments are congruent if they have the same length.
ii. Two angles are congruent if they have the same measure.
iii. Two squares are congruent if they have the same side length.
iv. Two rectangles are congruent if they have the same length and same breadth. v. Two circles are congruent if they have the same radius.

4. Two triangles are said to be congruent, if pairs of corresponding sides and the corresponding angles are equal.

If ΔABC and ΔPQR are congruent, then we write it as $\Delta ABC \cong \Delta PQR$.

5. The congruency of ΔABC and ΔPQR under the matching $\Delta ABC \leftrightarrow \Delta PQR$ is written as $\Delta ABC \cong \Delta PQR$.

This means the vertices A, B, C are matched with the vertices P, Q, R respectively.

6. SSS congruence condition: If three sides of a triangle are equal to the three sides of another triangle, then the two triangles are congruent.

7. SAS congruence condition: If two sides and the included angle of one triangle are respectively equal to the two sides and the included angle of another triangle, then the two triangles are congruent.

8. ASA congruence condition: If two angles and the included side of one triangle are respectively equal to the two angles and the included side of another triangle, then the two triangles are congruent.

9. RHS congruence condition: If the hypotenuse and a side of one right angled triangle are equal to the hypotenuse and a side of another right-angled triangle, then two triangles are congruent.

10. Two congruent figures are equal in area but two figures having the same area need not be congruent.

Congruent Figures

Congruent figures are exactly equal in size and shape.

In Geometry, congruence is a term used to define two objects that have same dimensions and shape. Moreover, if one shape can be exactly like the other on turning, flipping and/or sliding it, then the two shapes are said to be congruent to each other. So, two congruent figures drawn on a piece of paper can be cut out and placed over one another to match up perfectly. When we say a figure A is congruent to a figure B, symbolically it can be written as figure A ≅ figure B. In this article, we will discuss congruence of plane figures and line segments and congruent angles.

Congruence of Plane Figures

You are given two plane figures. Using the method of superposition, here are the steps to find whether they are congruent to each other or not:

Congruent Figures

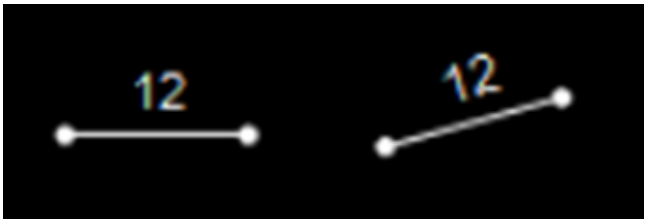
➥ Trace a copy of figure 1 using a tracing paper.

➥ Cut it out of the paper and place it over figure 2.

➥ You may slide or move or even turn over the paper. If they cover each other completely, they are congruent. We write figure 1 ≅ figure 2.

Congruence Among Line Segments

Two-line segments are said to be congruent to each other if they have equal lengths. They may or may not align at the same angle with an axis or position in plane. To find whether one line segment is congruent to the other or not, we use the same method of superposition as discussed above. Thus, if the two ends of the two line segments lie on one another, they are congruent.



Congruent Line Segments

Congruent Angles

If two angles have the same measure in degrees, they are congruent angles. The angles may or may not lie in the same position or orientation on plane. The method of superposition (as described above) can be used to check if given angles are congruent angles or not.

Congruent Angles

Thus, ∠PQR ≅ ∠ABC because the two angles measure 40 degrees. It is to be noted here that the length of BC is not equal to the length of QR. But congruence of angles does not depend on the length of the arms of the angles; it depends on the measures of the angles only. Thus, ∠PQR and ∠ABC are congruent angles.

Congruent Triangles

If all the sides and angles of a triangle are equal to the corresponding sides and angles of another triangle, then both the triangles are said to be congruent.

The meaning of congruent in Maths is addressed to those figures and shapes that can be repositioned or flipped to coincide with the other shapes. These shapes can be reflected to coincide with similar shapes.

Two shapes are congruent if they have the same shape and size. We can also say if two shapes are congruent, then the mirror image of one shape is same as the other.

In the above figure, Δ ABC and Δ PQR are congruent triangles. This means, Vertices: A and P, B and Q, and C and R are the same.

Sides: AB = PQ, QR = BC and AC = PR;

Angles: ∠A = ∠P, ∠B = ∠Q, and ∠C = ∠R.

Congruent triangles are triangles having corresponding sides and angles to be equal. Congruence is denoted by the symbol "≅". They have the same area and the same perimeter.

CPCT Full Form

CPCT is the term we come across when we learn about the congruent triangle. CPCT means "Corresponding Parts of Congruent Triangles". As we know that the corresponding parts of congruent triangles are equal. While dealing with the concepts related to triangles and solving questions, we often make use of the abbreviation cpct in short words instead of full form.

CPCT Rules in Maths

The full form of CPCT is Corresponding parts of Congruent triangles. Congruence can be predicted without actually measuring the sides and angles of a triangle. Different rules of congruency are as follows.

SSS (Side-Side-Side)

SAS (Side-Angle-Side)

ASA (Angle-Side-Angle)

AAS (Angle-Angle-Side)

RHS (Right Angle-Hypotenuse-Side)

Criteria for Congruency

SSS Criteria for Congruency

If under a given correspondence, the three sides of one triangle are equal to the three corresponding sides of another triangle, then the triangles are congruent.

Theorem: In two triangles, if the three sides of one triangle are equal to the corresponding three sides (SSS) of the other triangle, then the two triangles are congruent.

SAS Criteria for Congruency

If under a correspondence, two sides and the angle included between them of a triangle are equal to two corresponding sides and the angle included between them of another triangle, then the triangles are congruent.

ASA Criteria for Congruency

If under a correspondence, two angles and the included side of a triangle are equal to two corresponding angles and the included side of another triangle, then the triangles are congruent.

AAS Criteria for Congruency

AAS Rule: Triangles are congruent if two pairs of corresponding angles and a pair of opposite sides are equal in both triangles.

Why SSA and AAA congruency rules are not valid?

Two triangles with equal corresponding angles need not be congruent. In such a correspondence, one of them can be an enlarged copy of the other. Therefore, AAA congruency is not valid.

If two triangles have two congruent sides and a congruent non included angle, then triangles are not necessarily congruent. Therefore, SSA congruency is not valid.

RHS Criteria for Congruency

If under a correspondence, the hypotenuse and one side of a right-angled triangle are respectively equal to the hypotenuse and one side of another right-angled triangle, then the triangles are congruent.

Theorem: In two right-angled triangles, if the length of the hypotenuse and one side of one triangle, is equal to the length of the hypotenuse and corresponding side of the other triangle, then the two triangles are congruent.

Criteria for Congruency

Criteria for Congruency of two triangles are:

(i) SSS Rule

(ii) SAS Rule

(iii) ASA Rule

(iv) RHS Rule

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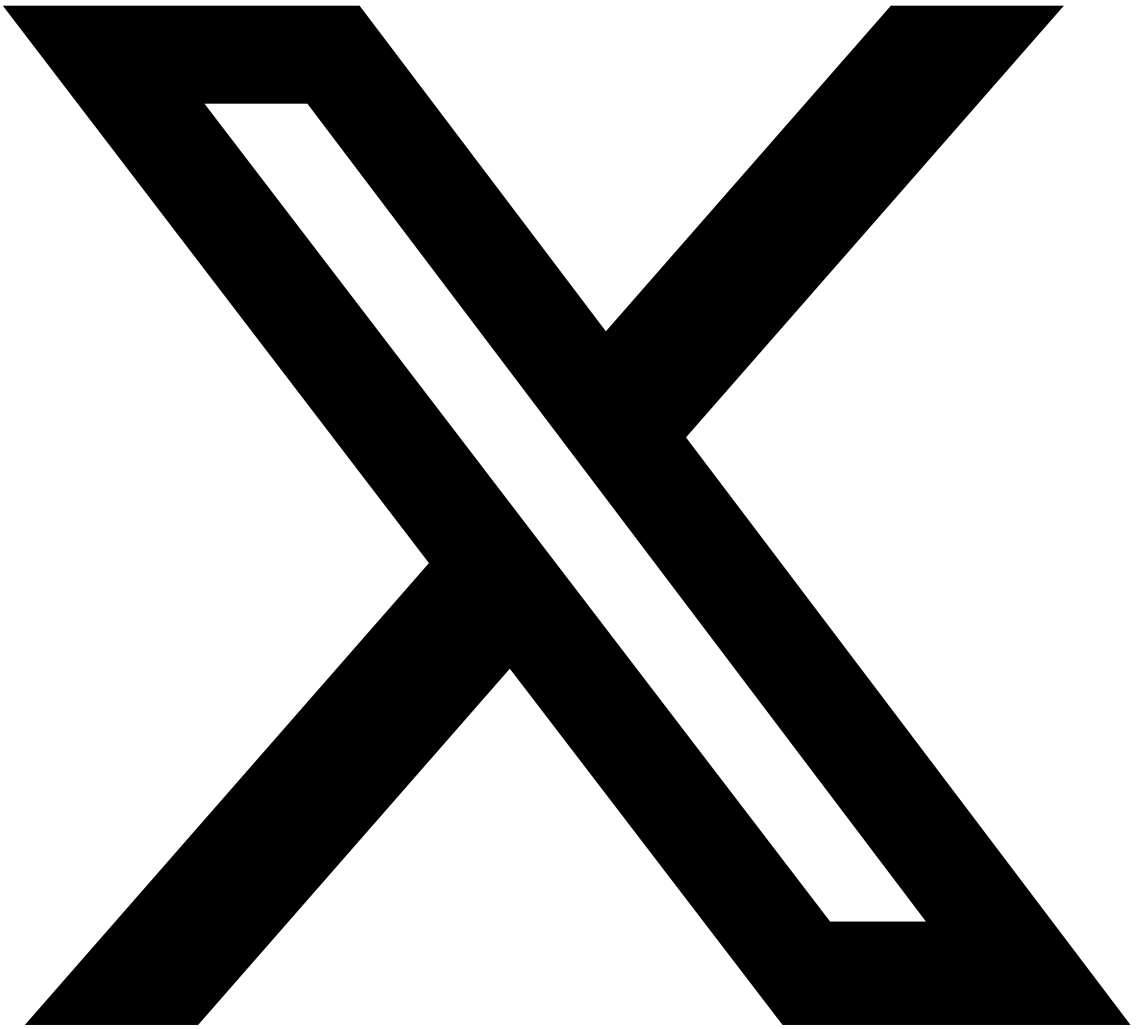

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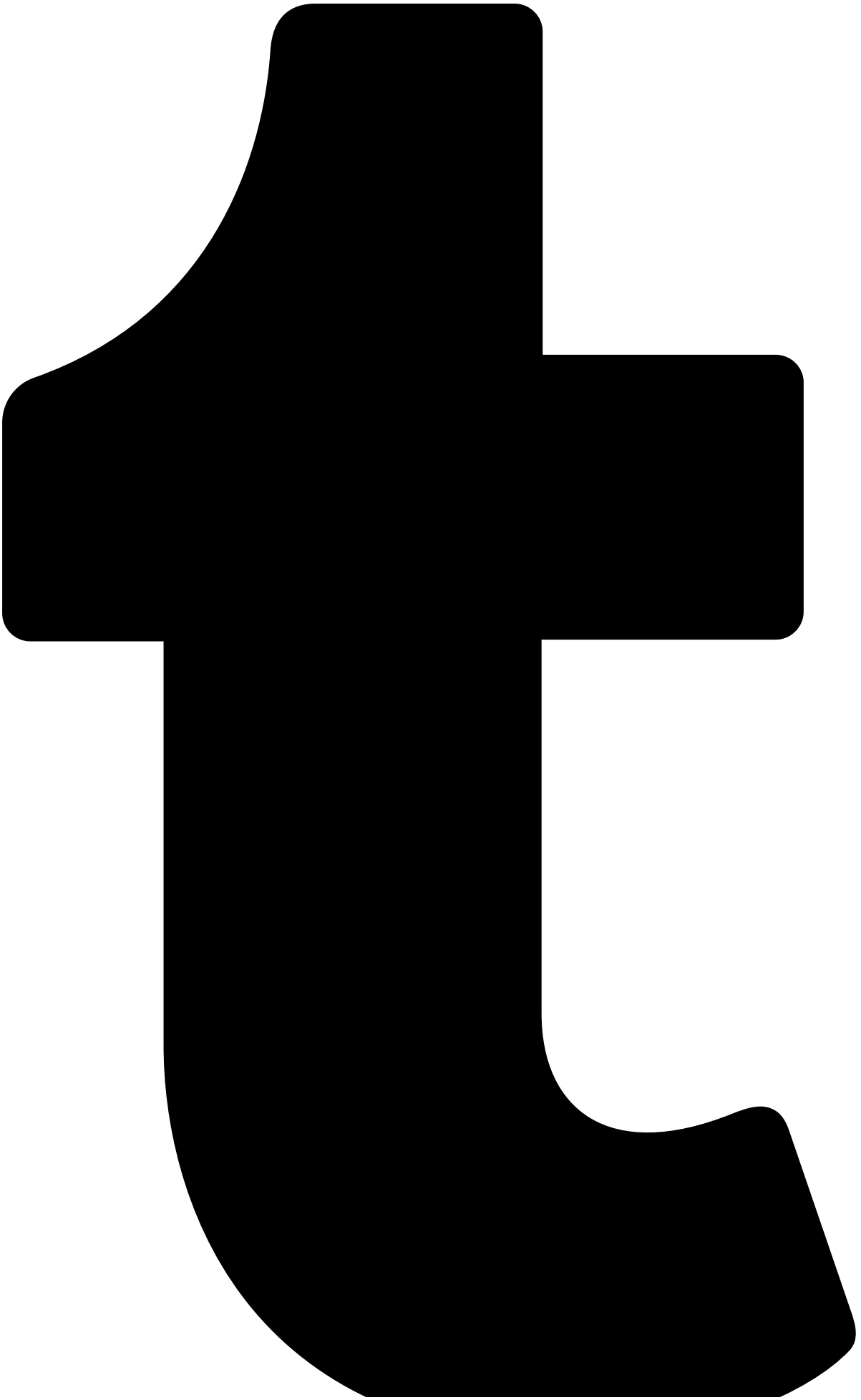

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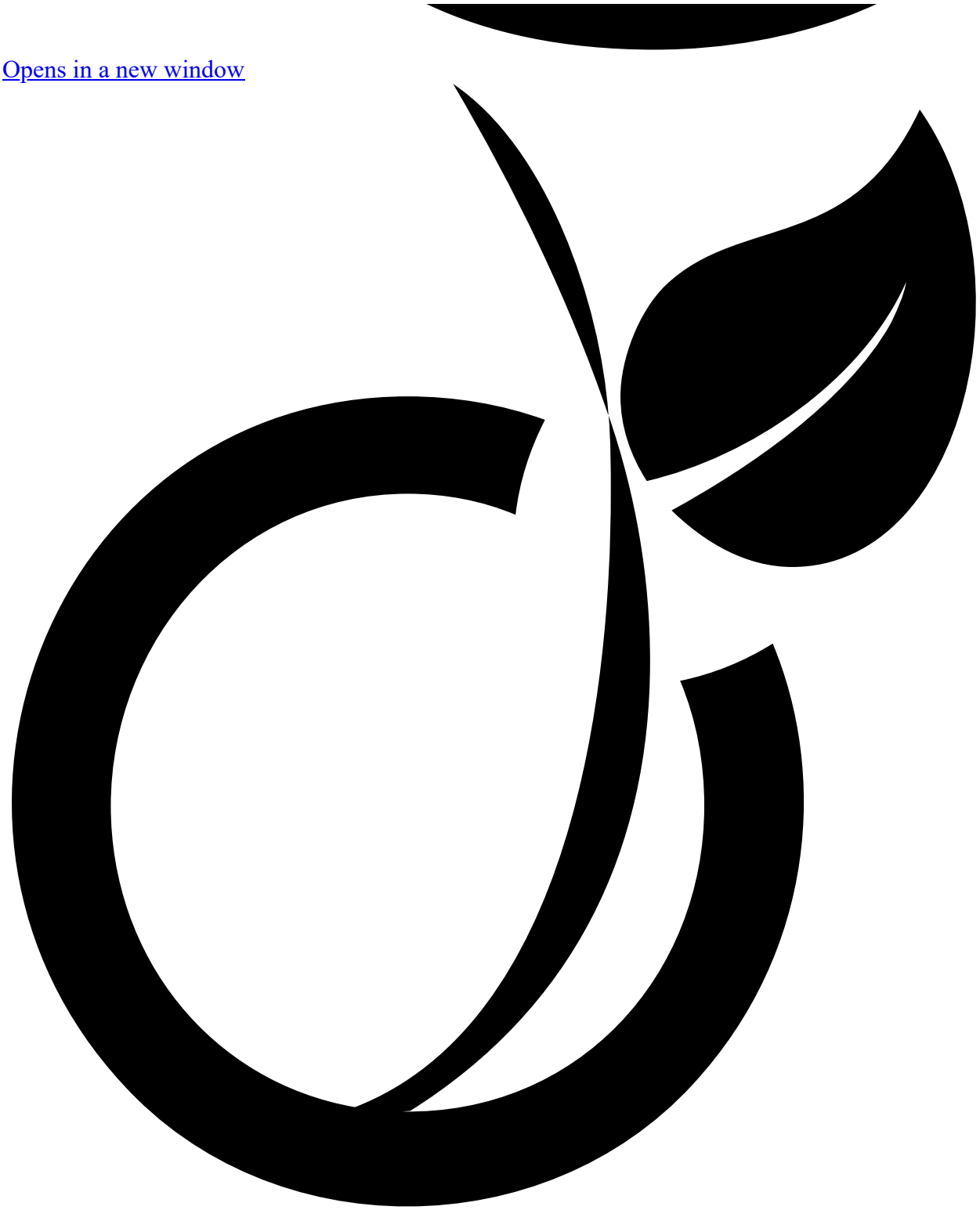

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